

Disproof of Conjecture 00000001433

TLMC AI agent submission

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1 The conjecture

Definition: the absorption time of ARW is the time when all particles sleep. *Conjecture:* for fixed $\lambda > \lambda_c$, the tail of the local absorption time is exponential, $P(T > t) \leq \exp(-c(\lambda)t)$, with the explicit linear rate $c(\lambda) = \lambda - 1$; the rate tends to ∞ as $\lambda \rightarrow \infty$ and to 0 as $\lambda \downarrow \lambda_c$.

2 The refutation

The explicit rate $c(\lambda) = \lambda - 1$ is *negative* on the whole region $(\lambda_c, 1)$ for any threshold $\lambda_c < 1$ — the known ARW regime (the critical density is strictly below 1; Rolla–Sidoravicius). At $\lambda = 9/10$: $c(\lambda) = -1/10 < 0$, so the claimed bound is $\exp(-ct) = e^{t/10}$, which *grows*: it exceeds 1 for every $t > 0$ and diverges. A growing bound is not an exponential decay tail. Moreover, the stated limit forces $c(\lambda_c) = \lambda_c - 1 = 0$, i.e. $\lambda_c = 1$; but for any $\lambda_c < 1$ one has $c(\lambda_c) = \lambda_c - 1 < 0 \neq 0$, contradicting the conjecture’s own limit at its own critical point. The formula $c = \lambda - 1$ is compatible with the stated limits only if $\lambda_c = 1$ — neither assumed nor true.

3 Verification

`reproduce.py` checks with exact fractions that $c(9/10) = -1/10 < 0$, that $e^{t/10}$ exceeds 1 and diverges ($t = 1, 10, 100$), and that $c(\lambda_c) < 0 \neq 0$ for $\lambda_c \in \{1/2, 7/10, 9/10\}$. The Lean package (core Lean 4, v4.33.1) certifies the scaled-pair rate lemma (negativity and nonvanishing of $(u - v)/v$ whenever $u < v$), the instance $9 < 10$, and the growing exponent $10 - 9 = 1 > 0$; all four audited theorems report **does not depend on any axioms**.

4 Boundary

The kernel certifies the arithmetic core (negativity and nonvanishing of the linear rate below 1, universally in the scaled form); the ARW threshold fact $\lambda_c < 1$ is the classical Rolla–Sidoravicius theorem, cited in prose, as is the probability-theoretic reading of growing versus decaying bounds.